

✓ 순전파 (Forward)

$$X \in \mathbb{R}^{m \times n}$$

$$Z_1 = XW_1 + b_1$$

$$A_1 = \sigma(Z_1)$$

$$Z_2 = A_1W_2 + b_2$$

$$A_2 = \sigma(Z_2)$$

$$Loss = -\frac{1}{m} \sum [y \log A_2 + (1 - y) \log(1 - A_2)]$$

✓ 역전파 (Backpropagation)

◆ 출력층

$$\frac{\partial Loss}{\partial A_2} = -\frac{y}{A_2} + \frac{1 - y}{1 - A_2}$$

$$\frac{\partial A_2}{\partial Z_2} = A_2(1 - A_2)$$

$$dZ_2 = \frac{\partial Loss}{\partial Z_2} = \frac{\partial Loss}{\partial A_2} \cdot \frac{\partial A_2}{\partial Z_2} = A_2 - y$$

◆ 출력층 파라미터

$$dW_2 = \frac{1}{m} A_1^\top dZ_2$$

$$db_2 = \frac{1}{m} \sum dZ_2$$

◆ 은닉층으로 오차 전파 (⚠ W_2 는 경로)

$$dA_1 = dZ_2 W_2^\top$$

◆ 은닉층 활성화함수

$$\frac{\partial A_1}{\partial Z_1} = A_1(1 - A_1)$$

$$dZ_1 = dA_1 \cdot A_1(1 - A_1)$$

◆ 입력층 파라미터

$$dW_1 = \frac{1}{m} X^\top dZ_1$$

$$db_1 = \frac{1}{m} \sum dZ_1$$

✅ 전체 흐름 요약 (그림용 한 줄)

$$Loss \longleftarrow dZ_2 \xleftarrow{W_2^\top} dA_1 \longleftarrow dZ_1 \longleftarrow (dW_1, b_1)$$

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